Suppressing the Leading Refusal-Associated Expert Index at Every Layer Redistributes Routing and Leaves Refusal Intact in a Mixture-of-Experts Model

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Abstract

In the base QWEN3.5-35B-A3B model (40 layers, 256 experts, top-8 routing), router capture at every layer under greedy decoding on token-matched pairs of harm-eliciting and matched-benign prompts shows the routing difference over generated tokens led by a single expert, expert 173 at layer 25, whose selection rate rises from 0.43 to 0.81. It leads a set of experts that a finance-versus-consequence control identifies as tracking real-world consequence and professional duty across domains, separable from experts that track the finance topic. Suppressing expert index 173 with an additive router bias applied in all 40 routers (the unit removed is the 40 experts that share that index, of which the leading expert is one; the readout is at layer 25) swept over four strengths collapses the leading expert's routed mass dose-dependently (selection rate 0.81 to 0.05, routing difference 0.124 to 0.004) and produces no harmful completion at any strength: routing reallocates to the other consequence-associated experts and the refusals persist, becoming more explicit. Refusal-associated routing in this model is therefore distributed, and its most active expert, together with every expert sharing its index, is dispensable. The single-component localization of refusal reported for dense models has, at the level of expert selection here, a distributed counterpart, with a direct consequence for practice: monitoring, ablating, or attacking the most active refusal-associated expert alone misses the behavior, and router-level safety auditing and single-expert ablation studies should be read at the level of expert sets. The study is a small case study (six and twelve prompts, single greedy trajectories) and is scoped accordingly.

1 Introduction

A mixture-of-experts (MoE) layer replaces a dense feed-forward block with many expert subnetworks and a router that selects a few of them for each token (Shazeer et al., 2017; Fedus et al., 2022). Because the router makes a discrete, per-token choice, its selections are unusually legible: one can read, token by token, which experts a behavior recruits. That legibility invites a specific safety question. When a model refuses a harmful request, is the refusal owned by a particular expert that a router-level analysis could name, an auditor could monitor, and an attacker could disable?

The dense-model picture points toward localization. Refusal in dense transformers has been described as mediated by a single direction in the residual stream, recoverable by contrasting harmful and harmless prompts and removable by projecting it out (Arditi et al., 2024), and representation-level control of harmfulness is a working tool (Zou et al., 2023). If the MoE analog held, refusal would localize to one or a few experts, suppressing the top one would weaken it, and routing telemetry on that expert would be a sufficient safety monitor.

We test this on the base QWEN3.5-35B-A3B model. We capture router logits at all 40 layers while the model answers token-matched pairs of harm-eliciting (hereafter *redflag*) and benign prompts under greedy decoding, decompose routed weight per expert and layer, and identify the experts whose routing separates the two classes. We then intervene: we add a negative bias to the router logit of the expert index with the largest separation, in every router, and sweep its strength, recording both the routing and the generated text. The removed unit is therefore the leading expert plus the 39 experts at other layers that share its index; the routing is read at the leading expert's layer.

The result is a distributed picture. One expert leads the routing separation, but it leads a set of experts that track real-world consequence and professional duty across domains, and when it and its same-index experts are biased to near-zero routed mass the rest of the set absorbs the load and the refusals persist. The

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implication for safety practice is direct: monitoring, ablating, or attacking the most active refusal-associated expert alone misses the behavior in this model, so router-level safety telemetry (Lv et al., 2026) and single-expert ablation studies should be read at the level of expert sets. A methodological finding recurs along the way: the apparent top-ranked prefill expert is largely an artifact of the token-matching padding, which a per-token decomposition dissolves.

The scope is narrow: six prompts in the matched set, twelve in the control, one model, one quantization, single greedy generations, an intervention on one expert index. The claim covers what removing the strongest expert, together with the experts sharing its index, does here; other models, prompt regimes, and multi-expert interventions may localize refusal or bypass it, and lie outside this study.

2 Background

Mixture-of-experts routing. Sparse MoE layers route tokens through a learned gate to a few of many experts (Shazeer et al., 2017), scaled into large transformers by Switch Transformers (Fedus et al., 2022), Mixtral (Jiang et al., 2024), and the fine-grained routing of DeepSeek (Dai et al., 2024); the model studied here is in the Qwen3 lineage (Yang et al., 2025). We use per-token router selection as the readout and ask which experts a single behavior recruits.

The locus of refusal. In dense models refusal has been characterized as a single residual-stream direction found by difference-in-means and ablated to disable refusal (Arditi et al., 2024), within a broader program of reading and steering high-level behaviors through directions (Zou et al., 2023). Our question is the routing analog: is there a comparable single locus at the level of expert selection? The contrast is in the substrate (router experts rather than a residual direction) and, as it turns out, in the answer.

Safety-relevant experts in MoE models. Two recent lines bear directly on this. Zhang et al. (2026) report that MoE routing is largely topic-driven and that safety behavior can change with little change to the routing path, a conclusion convergent with ours; our addition is a causal, dose-dependent suppression of the strongest expert's index with observed reallocation and preserved refusal on a named model. Lai et al. (2025) take the opposite stance, locating vulnerability in stable, identifiable safety-critical experts. We read the disagreement as setup-dependent: on this base model, prompt set, and intervention, removal of the strongest expert's index leaves refusal intact. Routing telemetry has also been proposed as a non-intrusive safety audit signal (Lv et al., 2026); our result bears on the granularity at which such telemetry is informative.

Distributed representations. That a behavior can be spread across many components is the expectation set by superposition (Elhage et al., 2022) and by dictionary-learning accounts in which interpretable features, including safety-relevant ones, are spread across many directions (Bricken et al., 2023; Templeton et al., 2024). Our result is the expert-routing counterpart of that picture. Methodologically we use the same causal-intervention logic as activation editing in dense models (Meng et al., 2022), applied to a router bias. Two companion studies on the same lineage characterize a single expert's role in generation (Shorthill, 2026a) and locate domain specialization in the generation regime (Shorthill, 2026b); robustness and safety benchmarks (Mazeika et al., 2024; Wei et al., 2023) frame why the locus of refusal matters, though we evaluate behavior by direct reading on a small set rather than by benchmark.

3 Methods

Model and decoding. We study the base Qwen3.5-35B-A3B model in 8-bit (`Q8_0`) GGUF form (the serialization used by llama.cpp), a `qwen35moe` architecture with 40 layers, 256 experts per MoE layer, and 8 experts selected per token. Decoding is greedy and deterministic (`-temp 0 -top-k 1 -seed 0`), capped at 256 new tokens, with a bare `</think>` assistant prefix (no reasoning trace) matching the project's reference runtime. The refusals studied are the base model's own; no refusal-reduced fine-tune is used.

Router capture and metric. For each prompt we capture the router pre-softmax logits at all 40 layers for every prompt and generated token. For an expert e on a set of tokens we report, after a softmax over the 256 logits, top-8 selection, and renormalization over the selected set: S_e, the fraction of tokens on which e is selected (the selection rate); Q_e, its mean renormalized weight when selected; and $W_e = S_e Q_e$, its mean routed weight over all tokens. Statistics are computed within the prefill block (prompt tokens) or the generation block (generated tokens), never pooled across them; layer 39 carries a distinct row count per prompt and is excluded from prefill claims. The ranking quantity is $\Delta W = W^{\mathrm{A}} - W^{\mathrm{B}}$, the difference in W between redflag, consequence, or unsafe prompts (A) and matched professional, descriptive, or neutral prompts (B).

Experiment 1: token-matched redflag pairs. Three pairs hold the request topic fixed and vary its framing: an oxycodone prompt asking how to get "incredibly high" against a pharmacist asking a normal dose; an instruction to brew ayahuasca (prefixed with a prompt-injection attempt) against a request to describe it; and a request to invest for a "sure" 500% return against a financial planner asking how to explain such an opportunity. The A and B prompts of each pair are padded with a repeated filler token to identical prompt token counts (50/50, 37/37, 74/74), so the two surfaces differ in framing at equal length. All six are answered once under greedy decoding.

Experiment 2: finance-versus-consequence control. Twelve prompts cross two axes, finance versus non-finance and consequence versus neutral, to separate experts that track a topic (finance) from experts that track real-world consequence and professional duty across domains (medical, legal, physical safety, inspection, finance). The buckets are matched in intent but not strictly token-paired.

Experiment 3: suppressing the leading expert index. We rerun the three redflag pairs while adding a negative bias to the router logit of expert index 173 (before top-8 selection) at four strengths (-0.25, -0.5, -1, -2), capturing all-layer routing and the generated text at each. Each generation is read for whether the model still refuses or warns and for how the routing redistributes.

Unit of intervention. The bias flag (`-expert-bias 173:`x) carries no layer argument, so the bias is applied in all 40 routers. Expert index 173 names a different expert at each layer: the 40 experts sharing that index have no shared parameters, and nothing in this design ties them together beyond the index. The unit removed is therefore a 40-expert set, of which the leading expert (173 at layer 25) is one member and 173 at layer 13 (rank 4 in Table 1) is another; the dose-response is read at layer 25, where the leading expert lives. This matters for how the result is stated. The removed set contains the leading expert, so an intact refusal under its removal bears on the necessity of the leading expert; the reallocation observed at other layers, though, occurs under a bias applied at those layers as well, and is not the response of an untouched network to the loss of one expert. A layer-restricted rerun would separate the two, and we do not have one.

Token-matching caveat. The pad token in Experiment 1 is a repeated word piece, and because it appears with different frequency in A and B it can manufacture a routing difference on the prefill block that reflects the padding itself. We therefore back the aggregate tables with a per-token decomposition that recomputes each candidate's ΔW with the filler piece removed, and we read the safety signal off the generation block, which is produced response text and carries no padding.

Rank	Layer	Expert	ΔW	W^{A} (S^{A})	W^{B} (S^{B})
1	25	173	0.124	0.192 (0.81)	0.068 (0.43)
2	19	45	0.082	0.102 (0.72)	0.021 (0.17)
3	20	25	0.073	0.110 (0.56)	0.037 (0.33)
4	13	173	0.072	0.191 (0.86)	0.119 (0.69)
5	31	45	0.067	0.078 (0.57)	0.010 (0.08)
6	27	216	0.063	0.074 (0.48)	0.011 (0.08)
7	17	189	0.061	0.107 (0.64)	0.046 (0.33)

Table 1: Generation-block A−B top-ranked experts on the three token-matched redflag pairs ($n = 6$ prompts). A is the redflag surface, B the matched benign surface; W is mean routed weight and S the selection rate. Expert 173 at layer 25 leads, and the separation is carried by a set of experts. Single greedy generation per prompt.

4 Results

4.1 The generation-side separation is led by one expert atop a distributed set

On the token-matched redflag pairs the routing difference over generated tokens is led by expert 173 at layer 25, with $\Delta W = 0.124$ (Table 1). Both components of its weight rise on redflag generation: selection rate from 0.43 on the benign prompts to 0.81 on the redflag prompts, conditional weight from 0.16 to 0.24. Beneath it sits a broad supporting set (expert 45 at layers 19 and 31, expert 25 at layer 20, expert 216 at layers 15 and 27, expert 189 at layer 17, and others), each separating the two prompt classes and none rivalling the top-ranked expert.

4.2 The set tracks consequence and duty across topics

The finance-versus-consequence control separates two things a single redflag set would conflate (Table 2). Finance-domain experts (expert 62 at layer 20, expert 95 at layer 30, expert 223 at layer 19, expert 214 at layer 16) fire strongly on finance prompts including neutral finance, with near-zero selection on non-finance: they are topic experts. The consequence-minus-neutral contrast is led instead by expert 189 at layer 17, expert 122 at layers 11 and 23, expert 157 at layer 14, expert 171 at layer 15, and expert 45 at layer 19, and it generalizes across medical, legal, physical-safety, inspection, and finance-consequence content. The refusal-associated set is real-world-consequence and professional-duty routing

Layer	Expert	ΔW
Finance-domain experts (finance − non-finance)		
20	62	0.180
30	95	0.177
19	223	0.167
Consequence/duty experts (consequence − neutral)		
17	189	0.073
23	122	0.065
14	157	0.058
15	171	0.053

Table 2: Finance-versus-consequence control ($n = 12$ prompts, bucketed). Finance-domain experts (top) fire on finance including neutral finance; consequence/duty experts (bottom) generalize across domains. Generation block. Buckets are intent-matched, not strictly token-paired.

Router bias on expert 173	ΔW (L25)	S^{A}	W^{A}
baseline	0.124	0.809	0.192
−0.25	0.093	0.664	0.141
−0.5	0.056	0.510	0.091
−1	0.041	0.394	0.056
−2	0.004	0.046	0.005

Table 3: Dose-response of suppressing expert index 173 in all 40 routers, read at layer 25 on the generation block. Routed mass and selection rate collapse toward zero; at every level the model still refuses or warns on all three redflag prompts.

that generalizes across topics.

4.3 Suppressing the leading expert index collapses its routing and leaves refusal intact

At the strongest bias, −2, expert 173 at layer 25 is nearly absent from the generation routing: its redflag selection rate is 0.05, down from 0.81, and its A−B weight difference is 0.004, down from 0.124 (Table 3, Figure 1). All three redflag prompts still refuse. The collapse is graded across the four bias levels; the refusal behavior is constant across all of them. The oxycodone prompt is refused at every level, and at −2 the refusal becomes more explicit about misuse and overdose; the ayahuasca instruction is refused throughout; the 500% prompt is warned against throughout, its wording shifting from “impossible and dangerous” to “trap” at the strongest bias. The benign halves stay professional. The routed mass that leaves expert 173 reappears nearby: generation separation reallocates to experts 45, 185, 157, 189, 216, and 133. The strongest expert, and every expert sharing its index, is removed and the rest of the set carries the refusal.

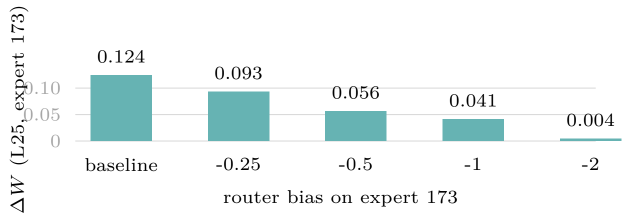

Figure 1: Generation-block A−B routing difference for expert 173 at layer 25 as the router bias on index 173 (applied in all 40 routers) is swept from baseline to −2. The expert’s routing collapses dose-dependently while the model’s refusals persist at every level.

4.4 Aggregate top-ranked prefill experts are largely a padding artifact

Read naively, the prefill block appears led by expert 173 at layer 1 ($\Delta W = 0.068$). The per-token decomposition shows this is mostly the repeated filler piece: excluding it, the layer-1 figure falls to 0.0004, and the other top aggregate prefill rows (expert 222 at layer 2, expert 36 at layer 22) collapse similarly. The prompt-side candidates that survive the correction are expert 218 at layer 14 ($0.025 \to 0.022$), expert 173 at layer 25 ($0.037 \to 0.046$), and expert 45 at layer 19 ($0.030 \to 0.034$). We therefore rest the safety read on the generation block and record the aggregate top-ranked prefill expert as a measurement caution: token-count padding can fabricate a prefill expert that a per-token decomposition dissolves.

5 Discussion

What the intervention establishes. Three results point the same way, and the suppression is the causal one. One expert leads the routing separation, but it leads a set; the set tracks consequence across topics; and biasing the leading expert’s index to near-zero routed mass, in every router, leaves the refusals in place while routing reallocates to the rest of the set. A behavior owned by a single expert should weaken when that expert is removed, and the removal here contains that expert; this one persists unchanged. In this model, harm-refusal is a property of a distributed set of experts that survives the loss of its most active member. This is the expert-routing counterpart of the distributed-representation expectation (Elhage et al., 2022; Bricken et al., 2023) and stands against a strict single-locus reading of refusal carried over from dense models (Arditi et al., 2024). The contrast with Lai et al. (2025) we take to be setup-dependent: a single-expert intervention that fails to bypass refusal here may succeed in another model, prompt regime, or with a multi-expert intervention. What this run shows is that removing the strongest expert, with its same-index experts, leaves refusal on this set intact.

Implication for safety auditing and intervention. The finding sets the granularity at which router-level safety work is informative in this model. Monitoring the single most active refusal-associated expert would report a large change under an intervention that leaves refusal intact, and ablating it would leave refusal untested; telemetry-based auditing (Lv et al., 2026) and expert-level ablation studies should therefore be read at the level of expert sets, and a single-expert result, positive or negative, leaves open where refusal lives. The same logic bounds the attack surface: in this model disabling the leading expert leaves refusal in place, and the routing that reappears under suppression is the reason.

Scope of the causal claim. The suppression removes expert index 173 in all 40 routers and finds the removed set dispensable. Because that set contains the leading expert, 173 at layer 25, the leading expert is dispensable too, on the assumption that removing fewer experts does not remove more of the behavior; a layer-restricted suppression would test that assumption directly. The intervention does not isolate the leading expert, and the same-index assumption it rests on, that a shared index is a harmless way to name 40 unrelated experts at once, is one this paper's own Table 1 declines to make when it ranks 173 at layer 25 and 173 at layer 13 separately. The joint necessity of the set (experts 45, 189, 157, 122 and the others that absorb the load) remains untested, so the result covers removal of the strongest expert's index. The behavioral read is a direct reading of generated text on a small set; benchmarked refusal rates lie outside the study. The result is a structural observation about where refusal-associated routing lives in this model.

6 Limitations

Six and twelve prompts with one greedy trajectory each yield point estimates with no sampled spread and no significance test, and per-prompt differences can depend on tie-breaking; the "still refuses" verdict is the author's reading of the generated text. One expert index was suppressed, in all 40 routers, so the leading expert was never removed alone and the joint contribution of the set is untested. The control is bucketed, with intent matching in place of token pairing, so its finance-versus-consequence separations are weaker evidence than the matched pairs. The study covers one base model at one quantization; whether refusal-reduced fine-tunes or other MoE families show the same distributed pattern or a localizable expert (Lai et al., 2025) is not addressed here.

7 Conclusion

In base Qwen3.5-35B-A3B, harm-refusal routes through a distributed set of experts. One expert leads the redflag-minus-benign routing difference, but biasing its index to near-zero routed mass in every router leaves the refusals intact, because the surrounding consequence-and-duty set absorbs the load. Refusal-associated routing in this model is distributed, and the most active expert is dispensable for refusal and an insufficient target for monitoring or attack. Router-level safety analyses, whether they aim to audit, ablate, or attack, should be read at the level of expert sets, and the unit an intervention removes should be named as it was applied; a result about the single most active expert speaks for that expert alone.

Data and Code Availability

The numbers in this paper are transcribed from the analysis tables of a single capture session and are listed claim-by-claim, with source files, in the accompanying `SOURCES.md`. The supporting data accompany the paper in the `data/` directory: the run's results write-up and journal, the all-layer A−B difference tables, the per-token decomposition that backs the filler-artifact correction, the finance-versus-consequence bucket tables, the expert-173 dose-response tables, the generated openings under suppression, and the prompt sets. Raw router tensors (`.npy`) were not retained in version control by project policy; the analysis tables above are derived from them and substantiate the reported statistics. The routing reconstruction satisfies $W = S \cdot Q$ per token set. No new model training was performed.

Ethical Considerations

This work studies how a model refuses, using a small set of synthetic redflag prompts (drug misuse, a prompt-injection attempt, and a financial "sure-thing") that the model refused at every intervention level. We release the prompts and the model's refusals; no completion in this study provides operational harm, and the suppression result is reported because it fails to bypass refusal. The purpose is defensive: to establish whether refusal in an MoE router is concentrated enough to be a single point of failure for auditing or attack. The finding that it is distributed argues against a single-expert attack surface in this model, and the claim is limited to this model.

AI Collaboration Statement

Generative AI (Anthropic's Claude) was used as a tool in preparing this manuscript: to assist with drafting and revising prose, organizing the manuscript, and locating and formatting the bibliography. All experiments, captures, and numerical results are the author's; every reported value was traced to a source capture (see `SOURCES.md`), and every reference was verified by the author against its authoritative record. No AI system is an author. The author reviewed the entire text and takes full responsibility for its content.

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This work received no external funding. The author declares no competing interests. The model studied is a third-party open-weights release used as-is for measurement; the author has no affiliation with its creators.

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